VSD

Option 09.2 - Inverter closed loop gain (15 – 125%) LV3

DEFAULT: 100
This option parameter affects all inverters connected to the control.

For normal operation, use 100%. If the inverter control is unstable, reducing this value has the effect of damping the inverter's response as its speed approaches its setpoint.

VSD

Option 09.3 - Inverter stop time (0 – 100 seconds) LV3

Determines the minimum time between a burner shutdown and subsequent startup. Set this parameter to give the inverter sufficient time to stop before the burner restarts. This parameter will increase the time the burner is held in status 5.

VSD

Option 09.4 - Inverter acceleration time (0–100 seconds) (60 seconds max. recommended) LV3

DEFAULT 30

The value entered for this parameter should be the time in seconds for the inverter(s) to move from minimum to maximum speed.

The inverter(s) should also be able to move from maximum to minimum speed in the same time, if this is not the case adjust the time in the inverter(s) to ensure the time to accelerate and de-accelerate are the same.

If this option parameter does not match the true VSD acceleration / deceleration times, drive positioning will be compromised and position faults will be likely.

Note: This parameter can be set to values between 60 and 100, but this should only be used as a last resort if required, and is not recommended. If a very slow inverter is used, care must be taken during commissioning that UP / DOWN keys are not held for more than a few seconds. It is possible that the control could get out of step with the inverter and cause a position fault during the commission process. This control is not designed to work with inverters that take more than 60 seconds to drive from zero to maximum speed.

VSD

Option 09.5 – VSD1 Speed Encoder Scaler (255 - 999) LV3

When using encoder feedback for VSD1, this option should be programmed to ensure that the feedback signal gives between 950 and 995 when the motor is at maximum speed (drive signal at 20mA). The value for the parameter may be calculated using the following formula:

$$\frac{\text{(Motor Max RPM x No of teeth on encoder)}}{60} = \text{Scaler}$$

The value may need adjustment once the unit has been tested. Specifically, it must be ensured that the feedback received never exceeds this value. In practice this may mean adding 2% to 5% to this value. See section 9.2for more details.

VSD

Option 09.6 – VSD2 Speed Encoder Scaler (255 - 999) LV3

When using encoder feedback for VSD2, this option should be programmed to ensure that the feedback signal gives between 950 and 995 when the motor is at maximum speed (drive signal at 20mA). The value for the parameter may be calculated using the following formula:

$$\frac{\text{Motor Max RPM x No of teeth on encoder}}{60} = \text{Scaler}$$

The value may need adjustment once the unit has been tested. Specifically, it must be ensured that the feedback received never exceeds this value. In practice this may mean adding 2% to 5% to this value. See section 9.2 for more details.

Option 09.7 – Reserved

Option 09.8 – Reserved

Option 14.0 – Primary Fault Relay (0 - 8) LV3
DEFAULT: Relay 4 (PD5)

This option parameter assigns the primary faults to a specific relay. Relays 2 & 3 share a common connection on the NX610 display. NXTSD104 relays are independent.

Option parameter 14.0 value	Meaning	Connection Details NX610
0	Primary faults not assigned to any relay.	N/A
1	Primary faults assigned to relay 1	Display, PR3(COM), PR1(NC), PR2(NO). Low voltage or line voltage.
2	Primary faults assigned to relay 2	Display, PR7(COM), PR5(NC), PR6(NO). Low voltage or line voltage.
3	Primary faults assigned to relay 3	Display, PR7(COM), PR9(NC), PR8(NO). Low voltage or line voltage.
4	Primary faults assigned to relay 4	PPC6000, PE1(COM), PE4(NO). Low voltage or line voltage.
5	Reserved for future expansion	N/A
6	Reserved for future expansion	N/A
7	Primary faults assigned to relay 7	Daughterboard, PZ15 – PZ16. LOW VOLTAGE AND CURRENT ONLY.
8	Primary faults assigned to relay 8	Daughterboard, PZ17 – PZ18. LOW VOLTAGE AND CURRENT ONLY.

These relays MUST NOT be used to provide a safety function.

A relay can be used to indicate any combination of fault conditions from 14.0, 14.1 and 14.2 – this means that one relay could be used for all faults. When used for an alarm function, the relay will de-energize when in the fault condition, so an alarm bell would be wired in series with the normally closed contacts.

Option 14.1 – Limit Relay (0 - 8) LV3

DEFAULT: Relay 4 (PD5)

This option parameter assigns the limits to a specific relay. Relays 2 & 3 share a common connection on the NX610 display. NXTSD104 relays are independent.

Option parameter 14.1 value	Meaning	Connection Details NX610
0	Limits not assigned to any relay.	N/A
1	Limits assigned to relay 1	Display, PR3(COM), PR1(NC), PR2(NO). Low voltage or line voltage.
2	Limits assigned to relay 2	Display, PR7(COM), PR5(NC), PR6(NO). Low voltage or line voltage.
3	Limits assigned to relay 3	Display, PR7(COM), PR9(NC), PR8(NO). Low voltage or line voltage.
4	Limits assigned to relay 4	PPC6000, PE1(COM), PE4(NO). Low voltage or line voltage.
5	Reserved for future expansion	N/A
6	Reserved for future expansion	N/A
7	Limits assigned to relay 7	Daughterboard, PZ15 – PZ16. LOW VOLTAGE AND CURRENT ONLY.
8	Limits assigned to relay 8	Daughterboard, PZ17 – PZ18. LOW VOLTAGE AND CURRENT ONLY.

These relays MUST NOT be used to provide a safety function.

A relay can be used to indicate any combination of fault conditions from 14.0, 14.1 and 14.2 – this means that one relay could be used for all faults. When used for an alarm function, the relay will de-energize when in the fault condition, so an alarm bell would be wired in series with the normally closed contacts.

Option 14.2 – Oxygen and Flue Temperature Limit Relay (0 - 8) LV3

DEFAULT: Relay 4 (PD5)

This option parameter assigns the limits to a specific relay. Relays 2 & 3 share a common connection on the NX610 display. NXTSD104 relays are independent.

Option parameter 14.2 value	Meaning	Connection Details NX610
0	Flue Limits not assigned to any relay.	N/A
1	Flue Limits assigned to relay 1	Display, PR3(COM), PR1(NC), PR2(NO). Low voltage or line voltage.
2	Flue Limits assigned to relay 2	Display, PR7(COM), PR5(NC), PR6(NO). Low voltage or line voltage.
3	Flue Limits assigned to relay 3	Display, PR7(COM), PR9(NC), PR8(NO). Low voltage or line voltage.
4	Flue Limits assigned to relay 4	PPC6000, PE1(COM), PE4(NO). Low voltage or line voltage.
5	Reserved for future expansion	N/A
6	Reserved for future expansion	N/A.
7	Flue Limits assigned to relay 7	Daughterboard, PZ15 – PZ16. LOW VOLTAGE AND CURRENT ONLY.
8	Flue Limits assigned to relay 8	Daughterboard, PZ17 – PZ18. LOW VOLTAGE AND CURRENT ONLY.

A relay can be used to indicate any combination of fault conditions from 14.0, 14.1 and 14.2 – this means that one relay could be used for all faults. When used for an alarm function, the relay will de-energize when in the fault condition, so an alarm bell would be wired in series with the normally closed contacts.

PPC Option 15.0 - Modulation sensor input type (0 - 3) LV3

The modulation sensor input caters for connection to a standard 0-5V, 4-20mA signal or a Fireye fail-safe pressure/temperature sensor. When using a Fireye fail safe sensor, the control may be used to monitor the boiler high safety limit and perform a non-volatile lockout if it is exceeded. Ensure that the links on the circuit board are set to correspond with the requirements of the sensor/signal being used.
If a 4.20mA sensor (Fireye PXMSxxx) is used it is still possible to set a safety limit, but an external limit device **must** be fitted to protect the boiler.

Option parameter 15.0 value	Meaning
0	0 - 5V operation. Set the JP1 link to 'OUT'. Set the JP3 link as required. This option is used to provide a 0 to 5 volt tracking input only. The burner modulation will track the voltage applied, going to high fire for 5 volts. There is no 'measured value', just a tracking setpoint. Option parameters 15.0 to 15.5 are unavailable.
1	4 - 20mA operation. Set the JP1 link to 'IN'. Set the JP3 link to'30V'. This option allows for connection to a 4 to 20mA pressure or temperature sensor. The burner modulation will track the current applied, going to high fire for 20mA and low fire for 4mA. If the current goes outside the range, the burner will go to low fire. There is no 'measured value', just a tracking setpoint. Option parameters 15.1 to 15.5 are unavailable.
2	4 - 20mA operation. Set the JP1 link to 'IN'. Set the JP3 link to'30V'. This option allows for connection to a 4-20mA measured value input device such as a pressure or temperature sensor. The internal PID will be used, if selected.

Option 15.1 – Modulation input decimal places (0 to 2)

This parameter specified the number of decimal places to which the measured value and setpoint are displayed. It also affects the scaling of the zero, span and safety limit – **it is vital that this parameter is set before parameters 15.2, 15,3 and 15.5**

Option parameter 15.1 value	Meaning
0	Measure value and setpoint displayed with no decimal places. Range of values is from 000 to 999.
1	Measure value and setpoint displayed with one decimal place. Range of values is from 00.0 to 99.9.
2	Measure value and setpoint displayed with two decimal places. Range of values is from 0.00 to 9.99.

PPC **Option 15.2 – Modulation input zero value (-999 to +999 / -99.9 to +99.9 / -9.99 to +9.99) LV3**

This value will normally be left at zero. It is the measured value to be displayed when the sensor connected is at its minimum value.

If a 4-20mA sensor is used, this parameter should be set to the 4mA value (usually zero).

NOTE: This option has been modified to allow for vacuum systems (available after June 2011).

PPC **Option 15.3 – Modulation input span value (-999 to +999 / -99.9 to +99.9 / -9.99 to +9.99) LV3**

This value is the measured value to be displayed when the sensor connected is at its maximum value.

If a 4-20mA sensor is used, this parameter should be set to the 20mA value.

NOTE: This option has been modified to allow for vacuum systems (available after June 2011).

PPC **Option 15.4 – Setpoint display units (0 – 3) LV3**

This option selects the displayed units for setpoint and measured value.

Option parameter 15.4 value	Meaning
0	Show measured value as 'PSI'.
1	Show measured value as 'bar'.
2	Show measured value as '°F'
3	Show measured value as '°C'
4	Show measured value as '%'
5	Show measured value as 'no units'

PPC **Option 15.5 - Boiler high safety limit (0 - 999 / 0.0 - 99.9 / 0.00 – 9.99) LV3**

If a 4-20mA sensor is used, and this parameter is set to a value other than zero, PPC6000 will *lockout* when the value is *exceeded*. ***Note: When a 4-20mA sensor is used, external limits must be in place to protect the boiler in case of sensor failure. Zero disables this parameter.***

Option 15.6 – Modulation Time (0 – 120 seconds) LV3

This option parameter sets the minimum time the burner will take to modulate from low to high fire or vice versa. Note - only the modulation speed in AUTO mode is affected. The burner may modulate slower than this setting if the drive speeds dictate this at any point in the range.

Option 15.7 – Bumpless Transfer (0 or 1) LV3

This parameter affects the burner operation while in MANUAL mode only.

Option parameter 15.7 value	Meaning
0	When the burner comes back on after going off, it will remain at low fire.
1	When the burner comes back on after going off, it will go to the last modulation rate that it was set to in manual mode and stay there.

Option 15.8 – Low before Off (0 or 1) LV3

When set to 1, this parameter will change the way a normal controlled shutdown works.

Option parameter 15.8 value	Meaning
0	The burner will turn off immediately when it is expected / required to.
1	The burner will modulate down for up to 30 seconds (or until low fire is reached) and then turn off.

Note: This function works for shutdowns caused by control limits for the currently selected setpoint and for shutdowns caused by option parameter 20.1 (aux shutdown) only. Lockouts / shutdowns caused by the alarm inputs in parameters 18.X will always work immediately.

Option 15.9 - *reserved*

Option 16.1 – Go back to pilot (0 to 15) LV3

This option allows a digital input to be configured to force the control to modulate down to low fire (if not already there) and then move to the ignition position (P2) and close the main fuel valves. When at P2 the ignition-prove output (LFS) comes on. The burner will continue to run with only the pilot on (in status 12) until the digital input is removed. The ignition transformer will not come on during the time that the 'go back to pilot' input is on, however it may come on briefly when the input is removed as part of the normal start-up procedure, as determined by option parameter 14.6.

This function can be used to prevent the burner from having to go off when the demand is low, meaning that it is ready to immediately respond to a sudden increase in demand (no pre-purge required).

The digital input number to use for this function is entered as the option parameter value.

NOTE: This option should only be used if the pilot is designed for continuous operation. Consult the burner manufacturer, national, state and local codes.

Number entered in parameter 16.1	Digital input used:
0	None.
1	Input 1, PA5 to PA11 Low Voltage supplied from PA11 **ONLY**
2	Input 2, PA6 to PA11 Low Voltage supplied from PA11 **ONLY**
3	Input 3, PA7 to PA11 Low Voltage supplied from PA11 **ONLY**
4	Input 4, PA8 to PA11 Low Voltage supplied from PA11 **ONLY**
5	Do not use.
6	Do not use.
7	Not a real input. Used for custom applications.
8	Not a real input. Used for custom applications
9	Not a real input. Used for custom applications
10	Not a real input. Used for custom applications
11	Not a real input. Used for custom applications
12	Not a real input. Used for custom applications
13	Not a real input. Used for custom applications
14	Not a real input. Used for custom applications
15	Not a real input. Used for custom applications

Option 16.2 – Allow profile swap (0 to 16) LV3

This option allows a digital input to be configured to allow a profile swap without turning the burner off. If this input is ON and a fuel profile selection change is made, the control will go to low fire then back to pilot ignition (P2) on the original profile. It will then close the main fuel valves and run with just the pilot on (and the ignition transformer if option parameter 14.6 is not set to 1). It will then move all drives to the P2 position of the new profile and open the appropriate main fuel valves.

The digital input number to use for this function is entered as the option parameter value.

Number entered in parameter 16.1	Digital input used:
0	None.
1	Input 1, PA5 to PA11 Low Voltage supplied from PA11 **ONLY**
2	Input 2, PA6 to PA11 Low Voltage supplied from PA11 **ONLY**
3	Input 3, PA7 to PA11 Low Voltage supplied from PA11 **ONLY**
4	Input 4, PA8 to PA11 Low Voltage supplied from PA11 **ONLY**
5	Do not use.
6	Do not use.
7	Not a real input. Used for custom applications.
8	Not a real input. Used for custom applications
9	Not a real input. Used for custom applications
10	Not a real input. Used for custom applications
11	Not a real input. Used for custom applications
12	Not a real input. Used for custom applications
13	Not a real input. Used for custom applications
14	Not a real input. Used for custom applications
15	Not a real input. Used for custom applications

Option 17.0 – Relay output function LV3

This option parameter assigns 'events' to the relay outputs. Set option parameter 17.1 to select the function for relay output 1, option 17.2 to select the function for relay 2 etc.

Option parameter 17.x value	Meaning
0	No function set from this option parameter
1	Digital Input 1 (PB9 to PB10 Low Voltage)
2	Digital Input 2 (PB9 to PB11 Low Voltage)
3	Digital Input 3 (PB9 to PB12 Low Voltage)
4	Digital Input 4 (PB9 to PB13 Low Voltage)
5	Digital Input 5 (PB14 to PB15 Low Voltage)
6	Digital Input 6 (PB14 to PB16 Low Voltage)
7	Digital Input 7, (PB14 to PB17 Low Voltage)
8	Digital Input 8, (PB5 to PB6 Low Voltage)
9	Digital Input 9, (PB5 to PB7 Low Voltage)
10	Profile 2 select (PB6 – PB8) ONLY
11	Profile 4 select (PB7 – PB8) ONLY
12	i/p 12, PE4 to live High Voltage
13	i/p 13, PE5 to live High Voltage
14	i/p Burner Select (PE6)
15	Air Flow On
16	Warming limit exceeded
17	Flame Detected
18	Boiler Below Control Limit
19	Gas Profile Selected (profile fires gas)
20	Oil Profile Selected (profile fires oil)
21	Controller in 'Lockout'
22	Burner is Shutdown from a Limit or Input event
23	Burner on/off from the keypad (only applicable to touchscreen with NX6100)
24	N/A
25	Burner modulating
26	VSD Run
27	Fan Run
28	Gas booster function (ON status 6 to 16 inclusive if profile selected fires gas)
29	Lockout or shutdown as selection 21 or 22.
30	Draft control relay (ON at status 15-18 inclusive)
31	Profile 1 selected – same as 8
32	Profile 3 selected – same as 9
33	Profile 2 selected – same as 10
34	Profile 4 selected – same as 11
35	Spare
36	Spare
37	Spare
38	Burner Available – **ON** unless locked out or held **OFF** by fault or burner select or keypad is off.
39	Purging – burner status = 8 (Firmware Version 1.204 and above)
40	Purge Complete – burner status = 9 (Firmware Version 1.204 and above)
41	Burner NOT turned off from keyboard / touchscreen.
42	Relay de-energized
43	Relay on if in normal mode (inverse of 23). Relay off if in local mode, or control powered off.
44	Relay on if control is powered up.

Option parameter 17.x value	Meaning
45	Relay ON status 2 onwards and during lockout post-purge (Fan on if burner on or waiting for burner select.)
46	Burner available. Relay on status 2 to 16 inclusive.
47	Low fire output.
48	High fire output.
49 – 100	+
101 – 199	Fault numbers 1 – 99 control the relay
200	Relay de-energized
201 - 250	EK101 – EK150 control the relay

Details of the relay connection details are shown below. Relays 2 & 3 on the display share a common connection and 4 & 5 on the PPC6000ler share a common connection. When used to indicate a fault or limit the N/O contacts will be closed when **NO** alarm is present, to ensure the alarm indication is fail-safe, therefore alarm devices should be wired to the normally closed contacts.

Abbreviations: Com.= Common, N.C. = normally closed, N.O. or N. Open = normally Open

A) For NX610 – 9-key keypad, if set as Alarm, no alarm present, at power up, relays change state as follows::

NOTE: No connection to terminal 4 on NX610 display

Option Parameter	Relay Output	ALARM Function Connection Detail for NX610 Only
17.1	1	Display, Low or Line voltage PR1 Normally Closed PR2 Normally Open PR3 Common
17.2	2	Display, Low or Line voltage. PR5 Normally Closed PR6 Normally Open PR7* Common
17.3	3	Display, Low or Line voltage PR7* Common PR8 Normally Open PR9 Normally Closed
17.4	4	PPC6000, PE1 PE4 Line voltage output ONLY
17.5	5	N/A
17.6	6	N/A
17.7	7	NXDBVSD, PZ15, PZ16, Low voltage <50V AC/DC, 200mA
17.8	8	NXDBVSD, PZ17, PZ18, Low voltage <50V AC/DC, 200mA
17.9	9	Available on NXTSD104 Touchscreen display **ONLY**

*** common to relays 2 and 3**

B) For NX610, if set as General Purpose relay, connections are as follows::

NOTE: You MUST recycle power to reset relay function first

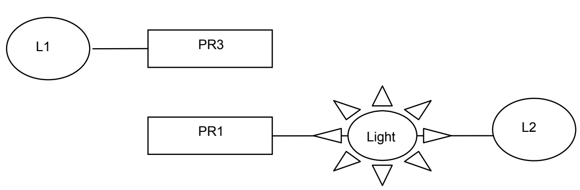
Option Parameter	Relay Output	GENERAL PURPOSE Connection Detail for NX610 Only
17.1	1	Display, Low or Line voltage PR1 Normally Open PR2 Normally Closed PR3 Common
17.2	2	Display, Low or Line voltage PR5 Normally Open PR6 Normally Closed PR7* Common
17.3	3	Display, Low or Line voltage. PR7* Common PR8 Normally Closed PR9 Normally Open
17.4	4	PPC6000, PE1 PE4 Line voltage output ONLY
17.5	5	N/A
17.6	6	N/A
17.7	7	NXDBVSD, PZ15, PZ16, Low voltage <50V AC/DC, 200mA
17.8	8	NXDBVSD, PZ17, PZ18, Low voltage <50V AC/DC, 200mA
17.9	9	Available on NXTSD104 Touchscreen display **ONLY**

*** common to relays 2 and 3**

For Example:

If you want an indicator light to illuminate when the warming limit is exceeded (released to modulate) -

1. Use one of the three display relays – for this example #1
2. Set 17.1 (for relay 1) to 16 (value for warming limit exceeded)
3. The contacts between PR3 and PR2 close when this is true.



Warming Limit Exceeded

Abbreviations: Com.= Common, N.C. = normally closed,
N.O. or N. Open = normally Open

C) For NXTSD104, if set as Alarm, no alarm present, at power up, relays change state as follows::